

FitBuddy – AI Fitness Plan Generator

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Planning Phase Overview

The Planning Phase defines how FitBuddy will be organized, developed and completed. It converts the project requirements into a practical development plan.

- Define objectives
- Identify modules and tasks
- Plan technologies and resources
- Plan schedule, testing and deployment

Project Objectives

The project is planned to generate personalized 7-day workout plans and nutrition/recovery tips using Gemini AI, while allowing feedback-based plan updates.

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Scope Planning

The planned system covers user input, AI workout generation, nutrition tips, feedback-based updates, database storage and an all-users view.

- In scope: workout generation
- In scope: nutrition guidance
- In scope: feedback and updating
- In scope: user/plan storage
- In scope: basic deployment

Expected Deliverables

The main deliverables are the source code, database, web pages, configuration files, test results and project documentation.

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Technology Planning

The project is planned using Python, FastAPI, Jinja2/HTML, SQLAlchemy with SQLite, Uvicorn and Google Gemini services.

- Frontend: HTML/Jinja2
- Backend: FastAPI
- AI: Gemini models
- Database: SQLite/SQLAlchemy
- Server: Uvicorn

Environment Planning

A Python virtual environment and a requirements file are planned so dependencies can be installed consistently.

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Task Breakdown

Development is divided into smaller tasks so each component can be implemented and tested separately.

- Set up project structure
- Configure Gemini API
- Develop AI generator functions
- Develop database functions
- Develop FastAPI routes
- Create templates
- Integrate and test

Responsibility Planning

Each module should have a clear responsibility. This reduces confusion and makes debugging easier.

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Schedule Planning

The project can be organized into sequential milestones: architecture, core functionality, route development, frontend, integration, testing and deployment.

- Milestone 1: architecture
- Milestone 2: core functionality
- Milestone 3: routes
- Milestone 4: frontend
- Milestone 5: integration/testing
- Milestone 6: deployment

Resource Planning

Required resources include a computer, Python environment, code editor, internet connection, Gemini API access and project documentation.

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Risk Planning

Potential risks include API errors, incorrect input, dependency problems, database errors and deployment configuration issues.

- Validate user input
- Keep API keys in environment configuration
- Handle AI failures
- Test database operations
- Maintain dependency list

Contingency Planning

If an individual module fails, it should be tested separately before integration. Backup copies of source code and database files should be maintained during development.

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Quality Planning

Quality is planned through modular coding, repeated testing, input validation and checking generated results.

- Test each function
- Test routes individually
- Check UI output
- Verify stored data
- Test feedback updates

Acceptance Planning

The project is considered ready when the main user flow works from input through plan generation, storage, feedback and updated-plan display.

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Deployment Planning

The application is planned to run with a Python environment, installed dependencies, configured Gemini API key and Uvicorn server.

- Prepare requirements
- Configure environment variables
- Run database setup
- Start Uvicorn
- Perform final checks

Maintenance Planning

After deployment, the project can be improved by refining prompts, improving interface design, fixing errors and adding new fitness features.

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Final Planning Summary

The Planning Phase provides a roadmap for completing FitBuddy in an organized manner. It identifies tasks, technologies, milestones, risks, testing and deployment activities.

- Clear objectives
- Defined modules
- Planned milestones
- Risk controls
- Testing strategy
- Deployment preparation

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Planning Conclusion

A structured plan reduces development confusion and provides a clear path from requirements to a working FitBuddy application.